

Exact Distribution, Matroid, and Lifting Addenda for the EG203 Obstruction Calculus

Jared Wilder

2026-05-13

Abstract

This addendum adds theorem-grade components to the EG203 stack. First, the local subgroup-obstruction indicators over a squarefree prime set are independent Bernoulli variables under CRT, giving an exact Poisson-binomial distribution and a direct Erdős–Kac normalization criterion. Second, Kummer log rows define a representable matroid over $\mathbb{F}_\ell$, with synchronization defects equal to matroid nullities. Third, prime-power lifting defects are identified with a precise generalized Wieferich relation map. Fourth, collision excess of Kummer slopes is expressed by a finite Fourier/Parseval identity over the rank- r Kummer Galois group. A final section records the Tate-module finite-linear analogue for elliptic and higher Galois representations.

1 Exact Poisson-binomial law for local obstruction counts

Let $\mathcal{Q}$ be a finite set of primes with $q \nmid ab$. Put

$$P = \prod_{q \in \mathcal{Q}} q, \quad G_P = (\mathbb{Z}/P\mathbb{Z})^*.$$

For each $q \in \mathcal{Q}$, write

$$\Gamma_q = \langle a, b \rangle \subset \mathbb{F}_q^*$$

and define the local membership indicator

$$B_q(m) = \mathbf{1}_{\Gamma_q}(-m^{-1}), \quad m \in G_P.$$

Let

$$p_q = \frac{|\Gamma_q|}{q-1}.$$

Theorem 1 (CRT independence of local obstruction indicators). *As m varies uniformly over G_P , the random variables*

$$\{B_q(m)\}_{q \in \mathcal{Q}}$$

are independent Bernoulli random variables with

$$\mathbb{P}(B_q = 1) = p_q.$$

Proof. By the Chinese remainder theorem,

$$G_P \simeq \prod_{q \in \mathcal{Q}} \mathbb{F}_q^*,$$

and uniform $m \in G_P$ corresponds to independent uniform residues $m_q \in \mathbb{F}_q^*$. The map $m_q \mapsto -m_q^{-1}$ is a bijection of $\mathbb{F}_q^*$, so

$$\mathbb{P}(B_q = 1) = \frac{|\Gamma_q|}{q-1} = p_q.$$

For any subset $S \subseteq \mathcal{Q}$,

$$\mathbb{P}(B_q = 1 \ \forall q \in S) = \prod_{q \in S} \frac{|\Gamma_q|}{q-1} = \prod_{q \in S} p_q,$$

again by CRT product measure. This is exactly independence. $\square$

Corollary 1 (Exact obstruction-count distribution). *Define*

$$N_{\mathcal{Q}}(m) = \sum_{q \in \mathcal{Q}} B_q(m).$$

Then $N_{\mathcal{Q}}$ has the Poisson-binomial distribution with generating function

$$\mathbb{E}z^{N_{\mathcal{Q}}} = \prod_{q \in \mathcal{Q}} (1 - p_q + p_q z).$$

Consequently,

$$\mathbb{E}N_{\mathcal{Q}} = \sum_{q \in \mathcal{Q}} p_q, \quad \text{Var}(N_{\mathcal{Q}}) = \sum_{q \in \mathcal{Q}} p_q(1 - p_q).$$

Proof. The generating function follows from independence:

$$\mathbb{E}z^{N_{\mathcal{Q}}} = \mathbb{E} \prod_q z^{B_q} = \prod_q \mathbb{E}z^{B_q} = \prod_q (1 - p_q + p_q z).$$

The mean and variance follow by differentiating the generating function, or directly by summing independent Bernoulli means and variances. $\square$

Theorem 2 (Erdős–Kac normalization criterion for obstruction counts). *Let $\mathcal{Q}_X$ be a sequence of finite prime sets and put*

$$V_X = \sum_{q \in \mathcal{Q}_X} p_q(1 - p_q).$$

If

$$V_X \rightarrow \infty \quad \text{and} \quad \frac{\max_{q \in \mathcal{Q}_X} p_q(1 - p_q)}{V_X} \rightarrow 0,$$

then

$$\frac{N_{\mathcal{Q}_X} - \sum_{q \in \mathcal{Q}_X} p_q}{\sqrt{V_X}}$$

converges in distribution to the standard normal law as $X \rightarrow \infty$.

Proof. By [Theorem 1](#), the summands $B_q - p_q$ are independent, centered, and bounded by 1 in absolute value. The displayed maximum condition is the Lindeberg condition for this triangular array, because for every fixed $\varepsilon > 0$, eventually

$$|B_q - p_q| \leq 1 < \varepsilon \sqrt{V_X}$$

for every q . The Lindeberg–Feller central limit theorem gives the conclusion. $\square$

Remark 1. *This theorem is exact finite CRT probability plus a standard triangular-array central limit theorem. Analytic number theory enters only when one estimates the subgroup densities $p_q = |\langle a, b \rangle \bmod q|/(q-1)$.*

2 The Kummer matroid

Let $r \geq 1$, let $\mathbf{a} = (a_1, \dots, a_r)$, and fix a prime ℓ . For primes $q \equiv 1 \pmod{\ell}$, $q \nmid a_1 \cdots a_r$, choose a primitive root $g_q \pmod{q}$ and write

$$a_j \equiv g_q^{u_{qj}} \pmod{q}.$$

Set

$$x_\ell(q) = (u_{q1}, \dots, u_{qr}) \in \mathbb{F}_\ell^r.$$

Definition 1 (Kummer log matroid). *For a finite prime set V , define $M_\ell(V; \mathbf{a})$ to be the representable matroid over $\mathbb{F}_\ell$ with ground set V and rank function*

$$\rho_\ell(S) = \text{rank}_{\mathbb{F}_\ell} \{x_\ell(q) : q \in S\}, \quad S \subseteq V.$$

The nullity is

$$\eta_\ell(S) = |S| - \rho_\ell(S).$$

Proposition 1 (Matroid axioms). *The function ρ_ℓ is a matroid rank function:*

$$0 \leq \rho_\ell(S) \leq |S|,$$

$$S \subseteq T \Rightarrow \rho_\ell(S) \leq \rho_\ell(T),$$

and

$$\rho_\ell(S \cup T) + \rho_\ell(S \cap T) \leq \rho_\ell(S) + \rho_\ell(T).$$

Proof. The first two properties are immediate for vector-space rank. For submodularity, let

$$U_S = \text{Span}_{\mathbb{F}_\ell} \{x_\ell(q) : q \in S\}, \quad U_T = \text{Span}_{\mathbb{F}_\ell} \{x_\ell(q) : q \in T\}.$$

Then

$$U_{S \cup T} = U_S + U_T,$$

and

$$U_{S \cap T} \subseteq U_S \cap U_T.$$

Therefore

$$\rho_\ell(S \cup T) = \dim(U_S + U_T) = \dim U_S + \dim U_T - \dim(U_S \cap U_T)$$

and since $\dim(U_S \cap U_T) \geq \dim U_{S \cap T}$, submodularity follows. $\square$

Theorem 3 (Synchronization defect equals matroid nullity). *For any finite $S \subseteq V$, the first-level ℓ -synchronization defect dimension of the Kummer log matrix on S is*

$$\delta_\ell(S; \mathbf{a}) = \eta_\ell(S) = |S| - \rho_\ell(S).$$

Proof. The first-level synchronization map is

$$\phi_{\ell, S} : \mathbb{F}_\ell^r \rightarrow \mathbb{F}_\ell^S, \quad n \mapsto (x_\ell(q) \cdot n)_{q \in S}.$$

Its matrix has rows $x_\ell(q)$. Hence

$$\text{rank}(\phi_{\ell, S}) = \rho_\ell(S).$$

Therefore

$$\dim \text{coker}(\phi_{\ell, S}) = |S| - \text{rank}(\phi_{\ell, S}) = |S| - \rho_\ell(S) = \eta_\ell(S).$$

$\square$

Definition 2 (Kummer Tutte polynomial). *The Kummer Tutte polynomial is*

$$T_{M_\ell}(x, y) = \sum_{S \subseteq V} (x-1)^{\rho_\ell(V) - \rho_\ell(S)} (y-1)^{|S| - \rho_\ell(S)}.$$

Corollary 2 (Tutte polynomial packages all first-level defects). *The exponent of $y-1$ in each term of T_{M_ℓ} is exactly the first-level synchronization defect dimension $\delta_\ell(S; \mathbf{a})$.*

Proof. By [Theorem 3](#), the exponent

$$|S| - \rho_\ell(S)$$

equals $\delta_\ell(S; \mathbf{a})$. □

3 Vertical lifting defect and generalized Wieferich relations

Let q be an odd prime and let $\mathbf{a} = (a_1, \dots, a_r)$ with $q \nmid a_1 \cdots a_r$. Put

$$\Gamma_{q^e}(\mathbf{a}) = \langle a_1, \dots, a_r \rangle \subset (\mathbb{Z}/q^e\mathbb{Z})^*.$$

The reduction map gives

$$1 \rightarrow K_e \rightarrow \Gamma_{q^{e+1}}(\mathbf{a}) \rightarrow \Gamma_{q^e}(\mathbf{a}) \rightarrow 1,$$

where

$$K_e = \Gamma_{q^{e+1}}(\mathbf{a}) \cap (1 + q^e\mathbb{Z}/q^{e+1}\mathbb{Z}), \quad |K_e| \in \{1, q\}.$$

Define the mod- q exponent-relation lattice

$$R_q(\mathbf{a}) = \{n = (n_1, \dots, n_r) \in \mathbb{Z}^r : a_1^{n_1} \cdots a_r^{n_r} \equiv 1 \pmod{q}\}.$$

For $n \in R_q(\mathbf{a})$, define

$$\lambda_q(n) = \frac{a_1^{n_1} \cdots a_r^{n_r} - 1}{q} \pmod{q}.$$

Lemma 1 (Well-defined homomorphism). *The map*

$$\lambda_q : R_q(\mathbf{a}) \rightarrow \mathbb{F}_q$$

is a group homomorphism from the additive relation lattice to the additive group $\mathbb{F}_q$.

Proof. For $n, m \in R_q(\mathbf{a})$, write

$$A(n) = a_1^{n_1} \cdots a_r^{n_r} = 1 + q\lambda_q(n) \pmod{q^2},$$

$$A(m) = a_1^{m_1} \cdots a_r^{m_r} = 1 + q\lambda_q(m) \pmod{q^2}.$$

Then

$$A(n+m) = A(n)A(m) \equiv 1 + q(\lambda_q(n) + \lambda_q(m)) \pmod{q^2}.$$

Thus $\lambda_q(n+m) = \lambda_q(n) + \lambda_q(m)$ in $\mathbb{F}_q$. □

Theorem 4 (Vertical lifting defect as generalized Wieferich relation). *At the first lifting level $q \rightarrow q^2$,*

$$K_1 \neq 1$$

if and only if λ_q is nonzero. Equivalently,

$$K_1 = 1$$

if and only if every mod- q exponent relation among the bases lifts to a mod- q^2 relation.

Proof. An element of K_1 is represented by a product

$$A(n) = a_1^{n_1} \cdots a_r^{n_r}$$

such that $A(n) \equiv 1 \pmod{q}$, but considered modulo q^2 it lies in

$$1 + q\mathbb{Z}/q^2\mathbb{Z}.$$

Thus $n \in R_q(\mathbf{a})$, and its image in the kernel is

$$A(n) \equiv 1 + q\lambda_q(n) \pmod{q^2}.$$

This kernel element is nontrivial exactly when $\lambda_q(n) \neq 0$. Therefore $K_1 \neq 1$ exactly when λ_q is not the zero homomorphism. $\square$

Corollary 3 (Classical Wieferich criterion, rank one). *Let $r = 1$, $\mathbf{a} = (a)$, and let $h = \text{ord}_q(a)$. Then*

$$K_1 = 1$$

if and only if

$$a^h \equiv 1 \pmod{q^2}.$$

Equivalently,

$$K_1 = 1$$

if and only if

$$a^{q-1} \equiv 1 \pmod{q^2}.$$

Proof. For $r = 1$, the relation lattice is generated by $h = \text{ord}_q(a)$. By [Theorem 4](#), $K_1 = 1$ exactly when

$$a^h \equiv 1 \pmod{q^2}.$$

Since $h \mid q-1$, write $q-1 = ht$, with $q \nmid t$. If $a^h \equiv 1 \pmod{q^2}$, then $a^{q-1} \equiv 1 \pmod{q^2}$. Conversely, if

$$a^h \equiv 1 + cq \pmod{q^2},$$

then

$$a^{q-1} = (a^h)^t \equiv 1 + tcq \pmod{q^2}.$$

Since $q \nmid t$, the congruence $a^{q-1} \equiv 1 \pmod{q^2}$ forces $c \equiv 0 \pmod{q}$. Hence $a^h \equiv 1 \pmod{q^2}$. $\square$

Definition 3 (Small Synchronization Defect lattice). *For a modulus vector $m_1, \dots, m_s$ and row vectors $u_i \in (\mathbb{Z}/m_i\mathbb{Z})^r$, define*

$$L(U) = \{n \in \mathbb{Z}^r : u_i \cdot n \equiv 0 \pmod{m_i} \text{ for all } i\}.$$

The Small Synchronization Defect Problem asks for a nonzero vector $n \in L(U)$ of minimal norm.

Proposition 2 (SSDP is a congruence-lattice shortest-vector problem). *The set $L(U)$ is a full-rank lattice in $\mathbb{Z}^r$. Solving SSDP is exactly solving the shortest nonzero vector problem in this explicitly given congruence lattice.*

Proof. Each condition

$$u_i \cdot n \equiv 0 \pmod{m_i}$$

defines the kernel of a homomorphism $\mathbb{Z}^r \rightarrow \mathbb{Z}/m_i\mathbb{Z}$. The intersection of finitely many finite-index sublattices of $\mathbb{Z}^r$ is again a finite-index full-rank sublattice. By definition, SSDP asks for a shortest nonzero vector in that lattice. $\square$

4 Collision excess and finite Fourier over the Kummer Galois group

Assume $a_1, \dots, a_r$ are distinct rational primes and set

$$K_{\ell, \star} = \mathbb{Q}(\zeta_\ell, a_1^{1/\ell}, \dots, a_r^{1/\ell}).$$

Over $\mathbb{Q}(\zeta_\ell)$,

$$G_\ell = \text{Gal}(K_{\ell, \star}/\mathbb{Q}(\zeta_\ell)) \simeq \mathbb{F}_\ell^r.$$

For unramified primes $q \equiv 1 \pmod{\ell}$, identify

$$\text{Frob}_q \in G_\ell$$

with the Kummer log vector $x_\ell(q) \in \mathbb{F}_\ell^r$.

Let $N : \mathbb{F}_\ell^r \rightarrow \mathbb{Z}_{\geq 0}$ be a finite count function; in applications

$$N(v) = \#\{q \leq Q : q \equiv 1 \pmod{\ell}, x_\ell(q) = v\}.$$

Let

$$\widehat{N}(A) = \sum_{v \in \mathbb{F}_\ell^r} N(v) e^{2\pi i A \cdot v / \ell}$$

for $A \in \mathbb{F}_\ell^r$, and put

$$\overline{N} = \ell^{-r} \sum_v N(v).$$

Theorem 5 (Collision-excess Parseval identity). *The collision excess*

$$\mathcal{C}_\ell(N) = \sum_{v \in \mathbb{F}_\ell^r} (N(v) - \overline{N})^2$$

satisfies

$$\mathcal{C}_\ell(N) = \ell^{-r} \sum_{\substack{A \in \mathbb{F}_\ell^r \\ A \neq 0}} |\widehat{N}(A)|^2.$$

Proof. Let

$$f(v) = N(v) - \overline{N}.$$

Then

$$\sum_v f(v) = 0,$$

so $\widehat{f}(0) = 0$ and $\widehat{f}(A) = \widehat{N}(A)$ for $A \neq 0$. Finite Fourier Parseval on the abelian group $\mathbb{F}_\ell^r$ gives

$$\sum_v |f(v)|^2 = \ell^{-r} \sum_A |\widehat{f}(A)|^2 = \ell^{-r} \sum_{A \neq 0} |\widehat{N}(A)|^2.$$

This is the displayed formula. □

Corollary 4 (Hyperplane subextensions control collision excess). *Each nonzero Fourier coefficient $\widehat{N}(A)$ is attached to the order- ℓ Kummer subextension*

$$K_{\ell,A} = \mathbb{Q}(\zeta_\ell, \alpha_A^{1/\ell}), \quad \alpha_A = \prod_{j=1}^r a_j^{A_j}.$$

Thus $\mathcal{C}_\ell(N)$ is exactly the average squared Frobenius bias over all nontrivial hyperplane Kummer subextensions.

Proof. The character

$$v \mapsto e^{2\pi i A \cdot v / \ell}$$

is the character of $G_\ell \simeq \mathbb{F}_\ell^r$ whose kernel is the hyperplane $A \cdot v = 0$. The corresponding quotient of G_ℓ has order ℓ , and its fixed field is

$$K_{\ell,A} = \mathbb{Q}(\zeta_\ell, \alpha_A^{1/\ell}).$$

The formula in [Theorem 5](#) expresses collision excess as the sum of squared biases over these nontrivial characters. $\square$

Remark 2. *This is the rigorous finite statement behind the Kummer-Vandiver/class-group probe direction. Any proposed connection to cyclotomic class groups must enter through unusually large Frobenius bias in the hyperplane subextensions $K_{\ell,A}$, and the statistic that sees that bias is $\mathcal{C}_\ell(N)$.*

5 Tate-module finite-linear obstruction calculus

The multiplicative Kummer rows $x_\ell(q) \in \mathbb{F}_\ell^r$ are one instance of a broader finite-linear obstruction pattern.

Let V be a finite-dimensional $\mathbb{F}_\ell$ -vector space and let $T_q \in \text{End}_{\mathbb{F}_\ell}(V)$ be a family of linear operators indexed by primes. For vectors $v_1, \dots, v_r \in V$, define the row map

$$\Psi_q : \mathbb{F}_\ell^r \rightarrow V, \quad (c_1, \dots, c_r) \mapsto (T_q - I) \left(\sum_{j=1}^r c_j v_j \right).$$

Definition 4 (Linear residual defect). *For a finite prime set S , define*

$$\Psi_S : \mathbb{F}_\ell^r \rightarrow V^S, \quad c \mapsto (\Psi_q(c))_{q \in S}.$$

The residual defect space is

$$D_{\ell,V}(S) = \text{coker}(\Psi_S).$$

Theorem 6 (Finite-linear obstruction identity). *For every finite S ,*

$$\dim_{\mathbb{F}_\ell} D_{\ell,V}(S) = |S| \dim V - \text{rank}_{\mathbb{F}_\ell} \Psi_S.$$

Proof. This is rank-nullity for the linear map

$$\Psi_S : \mathbb{F}_\ell^r \rightarrow V^S.$$

The codomain has dimension $|S| \dim V$. $\square$

Corollary 5 (Elliptic Tate-module shadow). *Let $E/\mathbb{Q}$ be an elliptic curve, let $V = E[\ell]$, and for primes $q \nmid \ell N_E$, let $T_q = \rho_{E,\ell}(\text{Frob}_q) \in \text{GL}(E[\ell])$. Then the residual defect space*

$$D_{\ell,E}(S) = \text{coker}(\mathbb{F}_\ell^r \rightarrow E[\ell]^S)$$

has dimension

$$\dim_{\mathbb{F}_\ell} D_{\ell,E}(S) = 2|S| - \text{rank } \Psi_S.$$

Proof. For an elliptic curve, $E[\ell]$ is a two-dimensional $\mathbb{F}_\ell$ -vector space. Apply [Theorem 6](#). □

Remark 3. *This proves the finite-linear Tate-module form of the obstruction calculus. Analytic distribution of Frobenius in the relevant Galois representation is a separate input.*

6 Stack insertion list

1. Add [Theorem 1–Theorem 2](#) to Paper 1 or as a new companion note on exact distribution laws.
2. Add [Definition 1–Corollary 2](#) to Paper 6.
3. Add [Theorem 4](#), [Corollary 3](#), and [Proposition 2](#) to the prime-power/Hensel section.
4. Add [Theorem 5–Corollary 4](#) to the Kummer slope collision graph program.
5. Add [Theorem 6–Corollary 5](#) as the finite-linear/Tate-module port.

References

- [1] P. Billingsley. *Probability and Measure*. Wiley, 3rd edition, 1995.
- [2] S. Lang. *Algebra*. Springer, revised 3rd edition, 2002.
- [3] L. C. Washington. *Introduction to Cyclotomic Fields*. Springer, 2nd edition, 1997.
- [4] J.-P. Serre. *Abelian l -adic Representations and Elliptic Curves*. Addison-Wesley, 1989.
- [5] J. Oxley. *Matroid Theory*. Oxford University Press, 2nd edition, 2011.
- [6] A. Wieferich. Zum letzten Fermatschen Theorem. *Journal für die reine und angewandte Mathematik*, 136:293–302, 1909.